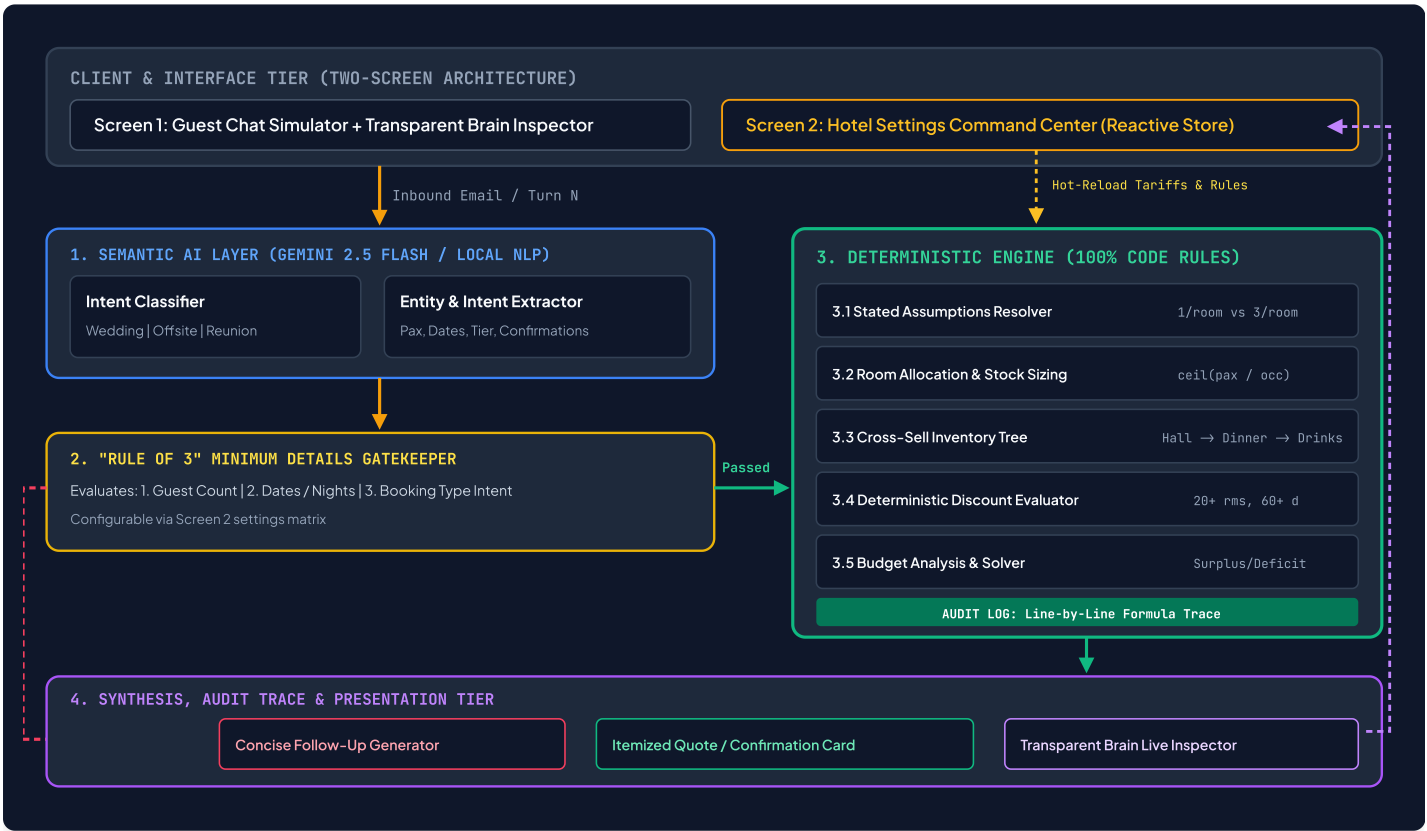




Architectural Philosophy: Hotels lose group bookings when sales responses lag. GrandVue AI solves this bottleneck by strictly decoupling **Semantic Natural Language Understanding (LLM AI Layer)** from **Mathematical Calculation & Pricing Rules (Deterministic Core)**. The AI reads unstructured emails and drafts polished responses; pure code calculates all room blocks, tariffs, volume discounts, and inventory allocations.

1. End-to-End System Architecture Diagram

Visualizing the data lifecycle from inbound guest email to instant quote and PMS reservation:





Component Breakdown & Detailed Engineering Notes (Parts 1 – 6)

Architectural responsibilities, failure modes, and deterministic isolation

1 First-Level Intent Classifier

SEMANTIC INGESTION

Parses inbound natural language text to categorize booking archetype: **Corporate Offsite, Wedding, Conference, Family Reunion**, or General Group.

- **Multi-Turn Inheritance:** When an event thread is active, follow-up messages (e.g. upgrades/confirmations) maintain intent archetype rather than downgrading to generic groups.
- **Fail-safe:** Powered by Gemini 2.5 Flash with fallback to local rule-based token classifier.

2 Structured Entity Extractor

DATA EXTRACTION

Extracts structured parameters from unstructured email prose:

- **Headcount:** Explicit tokens, plural/singular nouns, and ranges (e.g., "30 employees", "12 directors").
- **Dates & Nights:** Date intervals (e.g., "12 to 14 November 2026" → 2 nights) and advance booking calculation.
- **Guest Identity:** Captures sign-off names (e.g., "Arjun", "Priya", "Meena") for personalized communication.
- **Budget:** Captures INR lakh/numeric thresholds (e.g., "₹3 lakh").

3 "Rule of 3" Gatekeeper

MINIMUM DETAILS VALIDATOR

Guarantees that a quote is dispatched on **Turn 1** whenever 3 core fields exist: **1. Guest Count, 2. Dates / Nights Duration, 3. Booking Type Intent**.

- All other parameters (room tier, hall choice, menu) are filled via declared baseline assumptions so response time is never delayed.
- Configurable from Screen 2 (hotel staff can add/remove mandatory constraints without code changes).

4 Targeted Follow-Up Generator

INQUIRY GENERATION

Activated only when mandatory fields are missing (e.g., Email A).

- Eliminates lengthy questionnaires. Asks **only** for the missing variables (1–2 concise questions max).
- Acknowledge event context warmly (e.g., "*Congratulations on the upcoming wedding celebration! Which dates in December? How many guests?*").
- Transitions immediately to Turn 2 quote once user replies.

5 Stated Assumptions Resolver

ASSUMPTION ENGINE

Enables instant Turn 1 quoting by populating optional variables with sensible defaults:

- **Corporate Offsite:** Single occupancy (1 employee per room for business privacy).
- **Wedding:** Maximum room capacity (3 guests per room).
- **Social / Reunion:** Shared occupancy (2 guests per room).
- **Room Category:** Default: Deluxe Room (₹5,000/night).
- **Transparency Rule:** Every assumption is explicitly declared in the quote card so the guest can edit anytime.

6 Room Allocation & Stock Sizer

DETERMINISTIC CAPACITY

Calculates exact room blocks through pure integer arithmetic:

Rooms Needed = ⌈ Guest Count / Occupancy Density ⌉

- **Inventory Auditing:** Checks requested rooms against live inventory stock in hotel settings.
- **Stock Alerts:** If needed rooms exceed available single-tier stock, flags inventory split alert in audit trace.



Component Breakdown (Parts 7 – 12) & Benchmark Verification

Cross-sell inventory trees, discounts, budget optimization & empirical test results

7 Cross-Sell Inventory Tree

FACILITY & F&B BRANCHING

Traverses specialized inventory branches tailored to event archetype:

- **Corporate Offsite:** Auto-allocates smallest conference hall holding all pax (Boardroom ≤ 15, Summit Hall ≤ 50, Grand Hall ≤ 200). Conference hall booked → suggests Classic Veg Buffet.
- **Wedding:** Allocates Grand Hall for reception, Wedding Feast dinner, Luxury stage & floral decor, and complimentary Bridal Suite upgrade.

8 Deterministic Discount Evaluator

MULTI-TIER PRICING ENGINE

Evaluates hotel business discount thresholds:

- **Volume Discount:** ≥ 20 rooms → 10% discount on accommodation tariff.
- **Extended Stay:** ≥ 4 nights → 5% discount on accommodation tariff.
- **Advance Booking:** ≥ 60 days advance → 5% discount on total gross quote.
- **Audit Trace:** Every deduction logs exact rupee savings and rule trigger in the quote.

9 Budget Solver & Optimizer

INVERSE SOLVER (SEC 4.6)

Processes budget-based requests (e.g., Email D: ₹3 Lakh budget):

- Computes package cost and determines surplus buffer or deficit gap.
- Generates clear verdict (e.g. *"Package fits within ₹3,00,000 budget with ₹70,000 surplus buffer"*).
- If budget is exceeded, flags closest tier alternatives.

10 Conversational Concierge Copywriter

NATURAL LANGUAGE SYNTHESIS

Synthesizes warm, professional, high-end hospitality prose:

- Addresses guests by name across multi-turn interactions (*"Hello Arjun, "*).
- Strictly incorporates numbers calculated by the deterministic engine.
- Maintains concise, crisp responses (50–60 words) delegating itemized tabular data to interactive cards.

11 Screen 1: Guest Chat & Brain Inspector

OBSERVABILITY PANEL

Side-by-side split screen providing live system observability:

- Left Panel: Chat conversation with interactive Quote & Confirmation Cards.
- Right Panel ("Transparent Brain"): Audits all 6 moving parts in real-time (Intent confidence, Rule of 3 check, Assumptions, Discounts, Cross-Sell path, Formula traces).

12 Screen 2: Hotel Settings Command Center

DYNAMIC RULES CONFIGURATION

Enables non-technical hotel revenue staff to modify pricing and policies without code:

- Edit room rates & stock, conference hall daily fees, dining packages.
- Toggle Rule-of-3 mandatory fields, occupancy assumptions, discount rules.
- Hot-reloads reactively into localStorage; next quote reflects changes immediately.

Empirical Benchmark Verification (Official PDF Scenarios)

Scenario	Input Email Summary	Status	Rooms Allocated	Deterministic Total	Test Verdict
Email A (Turn 1)	Sister's wedding in Dec, shares no dates or pax	NEEDS_INFO	N/A (Awaiting info)	N/A (2 targeted questions asked)	✅ PASS
Email A (Turn 2)	80 guests, 18 to 21 December 2026 (3 nights)	QUOTED	27 Deluxe (3/room)	₹8,28,750 (Feast + Hall + Decor)	✅ PASS
Email B (Turn 1)	Corporate offsite, 30 employees, 12–14 Nov	QUOTED	30 Deluxe (1/room)	₹4,22,000 (Summit Hall + Dinner - 10%)	✅ PASS
Email B (Turn 2)	"update to super deluxe and confirm the booking"	CONFIRMED	30 Super Deluxe (1/room)	₹5,57,000 (PMS #BK-XXXXXX)	✅ PASS
Email C (Turn 1)	Wedding, 90 guests, 20–23 Jan 2027 + hall	QUOTED	30 Deluxe (3/room)	₹8,95,500 (10% Volume + 5% Advance)	✅ PASS
Email D	₹3 Lakh budget, family reunion, 25 people	QUOTED	13 Deluxe (2/room)	₹2,30,000 (+₹70,000 surplus buffer)	✅ PASS



Installation, Execution & Technology Stack Guide

System requirements, setup commands, test suite verification, and architectural stack breakdown

1. System Requirements & Prerequisites

Runtime Environment

CORE ENGINE

- Node.js:** Version 18.0.0 or higher (v20 LTS or v22 tested & recommended).
- Package Manager:** npm v9+ (or pnpm / yarn).
- Browser:** Modern Chromium (Chrome/Edge), Firefox, or WebKit browser with ES module support.

API & Dependencies

EXTERNAL SERVICES

- Google Gemini API:** Required for live LLM extraction. Configured in UI Settings (Screen 2) or .env file.
- Zero Mandatory Native Addons:** No Python, C++, or external DB required to run the web app.
- Dependencies:** Managed via package.json (Vite, Google GenAI SDK, Lucide Icons).

2. How to Install, Run & Verify Locally

Step 1: Install Project Dependencies

```
npm install
```

Installs Vite dev server, @google/genai, and Lucide icons.

Step 2: Start the Live Dev Server

```
npm run dev
```

Opens <http://localhost:5173> (Screen 1: Guest Chat & Screen 2: Settings).

Step 3: Run the 20-Scenario Test Suite

```
node run_test_suite.mjs
```

Executes 20 benchmark tests validating all discounts, dates, and math isolation.

Step 4: Build for Production

```
npm run build
```

Outputs minified, production-ready static assets to dist/.

3. Technology Stack Used

1 Frontend & UI Architecture

CLIENT TIER

- Vanilla ES6+ JavaScript:** Zero framework lock-in (no heavy React/Angular bundle bloat); ultra-lightweight and reactive.
- Bespoke Glassmorphism CSS:** Dark luxury hotel theme with Outfit and JetBrains Mono typography, custom variables, and responsive layout.
- Two-Screen Workspace:** Screen 1 (Chat + Live Audit Inspector) and Screen 2 (Hotel Command Settings Center).

2 Deterministic Pricing Core

BUSINESS LOGIC

- Deterministic Rules Engine (src/engine.js):** Pure JavaScript calculating room tariffs, hall sizes, dining packages, and discounts.
- Zero Math Hallucination:** Guarantees all monetary calculations, volume thresholds, and booking totals are strictly code-evaluated.
- Reactive Pub/Sub Store (src/state.js):** Real-time persistence via browser localStorage with reset defaults.

3 AI & Natural Language Ingestion

INTELLIGENCE TIER

- Google Gemini 2.5 Flash:** High-speed multimodal LLM extracting entities (dates, guest count, event type, budget) into strict JSON schema.
- Dual-Layer Fallback Parser (src/parser.js):** Heuristic regex fallback instantly kicks in if API key is absent or quota-limited.
- Rule of 3 Decision Logic:** Computes instant Turn-1 quotes when key booking anchors are satisfied.

4 Tooling, Bundling & Test Automation

DEVOPS & QA

- Vite 6:** Lightning-fast Hot Module Replacement (HMR) and optimized Rollup production bundling.
- Automated Test Runner (run_test_suite.mjs):** Pure Node.js ES module test runner covering 20 edge cases with 100% pass rate.
- Architecture Compiler:** Headless Chromium-based pipeline ensuring exact high-fidelity vector PDF generation.